INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 3: Research Methodology

by

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3 Research Methodology

Chapter 2 ended with five gaps. This chapter gives each a measurable procedure and sets out how the coupled walk C-PR, the method of the thesis, and its two single-layer references, EndorseRank and Adaptive Weighted PageRank (AWP), are evaluated as scores that should find the contracts of Arbitrum One that users rely on. EndorseRank runs on the ERC-20 allowance graph and AWP on the transfer graph (T. Do et al., 2023); C-PR walks both in one chain (Section 3.6.3). Every walk is solved with weighted PageRank (Page et al., 1999).

The evaluation has three parts: the discovery-order test, which asks whether a crawl ordered by a score reaches the contracts with the most user-account demand sooner than simpler orderings (Section 3.7); the out-of-window prediction of later approvals and inflow (Section 3.8); and the same-window evaluation, which follows construct (domain) alignment: each ranking method is one indicator of a reputation construct that cannot be observed directly, in the sense of construct validity (Cronbach & Meehl, 1955). The claim is comparative throughout: every score is set against the raw counts it smooths.

Four principles guide each operational choice.

- Same-seed comparability: every score is computed on an identical set of contracts, so no difference between scores can come from sampling.
- Fixing choices in advance: the analysis plan and the versioned configuration were committed to the authors' Git repository before the study extracted any log (Section 3.14). They fix the windows, the cohort and account-type rules, every score with its parameters, the values of λ, the proxies, the W0 labels, the bootstrap, the benchmark protocol, the robustness checks and rule A, the plan's one decision rule, whose object is C-PR (RQ4; Section 3.8). Scan A, the cohort, the ego extraction and the same-window and W0 results followed. The registration of W1 was committed next, with rule B and its margin (Section 3.8.1); scan B and the rest

of the registered analysis came after it (Appendix 8.6). Chapter 4 reports the main analysis, whose definitions Section 3.5.3 lists with their status. RQ4 and RQ5 follow the rules of the analysis plan and of the registration, applied to the scores of the contract graph as a re-analysis; RQ2 and RQ3 follow criteria defined after the registered results were known. Every contrast outside the rules was computed after its data were known and is reported as such, with its interval.

- Construction and timing: each cohort's edge lists are built once, before any timer starts, and the benchmarks time the solver call.
- Scope: RQ1, with H1 to H3 and Claims C5 and C6, concerns the matched cohort in the same window and the cost of each walk on the graphs of T_{obs}. RQ2 and Claim C1 concern crawls of the contract graph at each freeze. RQ3 to RQ5 and Claims C2 to C4 draw on the spender cohorts of W0 and W1, with the same-window part of rule A on the matched cohort.

3.1 Notation

The notation used throughout the thesis is listed in Table II; Table 3.1 adds the symbols that only the equations of this chapter use. Vectors are set in bold and sets in calligraphic type.
Except where stated otherwise, PageRank means the weighted variant of Page et al. (1999), whose update at iteration t is

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\left(\mathbf{r}^{(t)\top}\mathbf{1}_{\text{dang}}\right)\boldsymbol{\pi}, \tag{3.1}$$

where P is the transition matrix of Equation (3.2) below, $\boldsymbol{\pi}$ is the restart distribution over the nodes, uniform ($\pi_u = 1/|V|$) unless a method specifies another (AWP in Section 3.6.1, EndorseRank with activity restarts in Section 3.6.2, S-PR in Section 3.6.3), and $\mathbf{1}_{\text{dang}}$ marks the dangling nodes (Table 3.1). With $w(u,v)$ denoting the weight of the directed edge from u to v and $o(u) = \sum_v w(u,v)$

Table 3.1: Notation of this chapter's equations beyond the list of Table II.

Symbol	Meaning
$G = (V, E)$	Graph passed to the PageRank solver: every node is a contract, and V holds the end points of the retained edges E
V_A, E_A; V_T, E_T	Nodes and edges of the EndorseRank allowance graph $G_{\text{approve}} = (V_A, E_A, w_A)$ and of the AWP transfer graph $G_{\text{transfer}} = (V_T, E_T, w_T)$
$a^\star(u, v, \text{token})$	Latest in-window allowance of owner u to spender v on one token (US dollars)
$\text{bal}(u, \text{token})$	Balance of owner u on the token at the freeze (US dollars)
$\hat{a}(u, v, \text{token})$	Capped allowance, $\min(a^\star, \text{bal})$
$\Delta t^\star(u, v, \text{token})$	Age of the approval that set $a^\star$
$w_A(u, v)$	EndorseRank edge weight, owner $u \to$ spender v: $\sigma(\Delta t^\star)V(\hat{a})$ summed over tokens
$w_T(u, v)$	AWP edge weight, sender $u \to$ receiver v: $\sigma(\Delta t_e)V(x_e)$ summed over transfers
x_e	Amount of transfer e (US dollars)
b_T, b_A	Slopes of the value transform V for transfers and for allowances, $b = \ln 3/m$ with $m = m_T$ and $m = m_A$
X_u^A	Activeness of contract u in the allowance graph, the restart weight of EndorseRank with activity restarts
$\tilde{w}_A(u, v)$, $\tilde{w}_T(u, v)$	Weights of the two layers of C-PR, in US dollars
$D_\lambda(u)$	Common denominator of the layer weights $\alpha_A(u)$ and $\alpha_T(u)$ in node u's C-PR row
$\mathbf{1}[\cdot]$	Indicator: 1 if the condition holds, 0 otherwise
$\mathbf{1}_{\text{dang}}$	Indicator vector of the dangling nodes, entries $\mathbf{1}[o(u) = 0]$
$\mathbf{s}_M$, $\mathbf{p}_c$	Score vector of method M and value vector of check c over the cohort
R_i, S_i	Mid-ranks of contract i's score and proxy value (tied values share their average position)
n_c, n_d; n_0, n_1, n_2	Concordant and discordant contract pairs; all pairs, $n(n-1)/2$, and the pairs tied in the first and in the second ranking

the out-strength, the transition matrix is

$$P_{u,v} = \begin{cases} w(u, v)/o(u) & \text{if } o(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

The rows of dangling nodes are zero, so P is substochastic; the mass lost at those nodes returns through the third term of Equation (3.1) and is spread according to $\boldsymbol{\pi}$. Filling each zero row with

$1/|V|$ gives the row-stochastic matrix of Equation (2.1), and with uniform $\boldsymbol{\pi}$ Equation (3.1) is then the Google-matrix form of Equation (2.2). The iteration starts from $\mathbf{r}^{(0)} = \mathbf{1}/|V|$ and stops once $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or after $I_{\max}$ iterations, and the final vector is normalized to sum to one over V.

The default damping of 0.85 is the classical web setting (Page et al., 1999), and the robustness sweep varies it with edge weights held fixed (Section 3.13). The strict $\varepsilon = 10^{-8}$ and $I_{\max} = 300$ keep the measured runtimes tied to the structure of the graph and not to early termination.

3.2 Data sources and sampling

Figure 3.1 places every kind of on-chain record used in the thesis on one calendar.

- Primary chain: All events come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum (Thibault et al., 2022) whose logs are in the public BigQuery export. One chain avoids cross-chain aliases and gives every address the same fees and latencies.
- Extraction and score windows: Scan A extracts the events from 2023-10-01 00:00:00 UTC to 2026-06-30 23:59:59 UTC, and scan B those of the label window W1, July to September 2026, so the data cover three years. Every score of the main analysis is computed from the twelve months before its freeze: from 2025-07-01 00:00:00 UTC to T_{obs} = 2026-06-30 23:59:59 UTC in the same-window evaluation and in W1, and from 2025-04-01 00:00:00 UTC to t_1 = 2026-03-31 23:59:59 UTC in W0 (Section 3.8). A window of one year bounds the age of every event that enters a score, a proxy, a baseline or a label history and gives the two freezes windows of the same length. The time decay is anchored at the freeze, so an event on the first day of a window is 365 days old. The registered analysis (Appendix 8.6) scores every event of scan A from 2023-10-01.
- BigQuery source: Logs are read from the public BigQuery table

  `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

  Scans A and B copy the events the study needs into its own Google Cloud project, and later

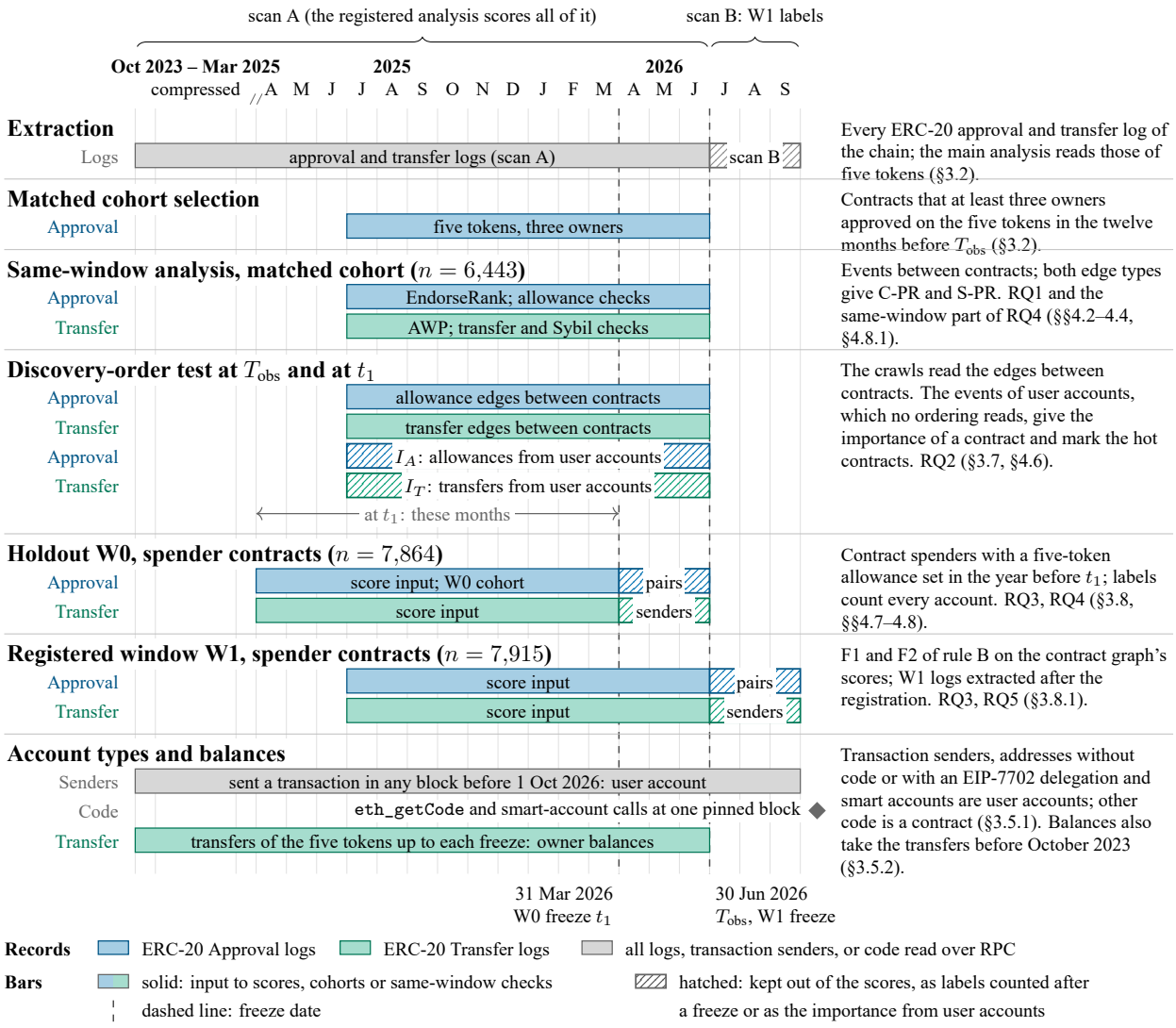


Figure 3.1: Blockchain data used in this thesis, all from Arbitrum One. Each lane is one analysis, and each bar is one kind of on-chain record placed on the months it covers; the 18 months before April 2025 are drawn at a compressed scale. Solid bars are inputs to scores, to cohorts or to same-window checks. Hatched bars are kept out of the scores: the labels counted after a freeze date and, in the discovery-order test, the events of user accounts that give the importance of a contract. The July–September logs were extracted only after the registration of W1 was committed. The approval logs of five tokens in the twelve months before each freeze select the cohort of that freeze; the graphs of the main analysis hold the events of those tokens between two contracts in those months, read from the whole chain; the registered analysis scores every event of scan A.

queries read those copies (Section 3.3); the same public dataset gives the addresses that sent a transaction and the transfers before scan A that enter the balances. Appendix 8.2 shows the query patterns, which anyone with a Google Cloud account can rerun.

- Reputation events: EndorseRank uses ERC-20 `Approval` logs (Vogelsteller & Buterin, 2015), which also cover approvals made through EIP-2612 `permit` (Lundfall, 2020), and AWP uses `Transfer` logs. Both are kept only for logs with exactly three topics and for the five tokens of Section 3.5.3. The main analysis reads every such `Approval` and non-zero `Transfer` of the chain, not only those around a cohort, from 1 April 2025 to 30 September 2026 (the typing window).
- Nodes and scored addresses: Every node of every graph is a contract, and only contracts are scored. A user account is an address that sent a transaction, an address without code, an EIP-7702 account, which an externally owned account (EOA) becomes by setting a delegation while keeping its key (Buterin et al., 2024), or a smart account (Section 3.5.1), that is a Safe (Safe Ecosystem Foundation, n.d.) or an ERC-4337 account (Buterin et al., 2021). No user account is a node, and no score or baseline reads an event of one: an EOA costs no deployment, and the study does not measure who controls a user account, whereas a contract costs a deployment and acts only as its code states. What user accounts do is kept apart, as the demand that the discovery-order test measures and that the labels count. The graphs mix contracts of every role and no analysis is stratified by role, so differences between roles enter every coefficient.
- Matched cohort: Alignment is applied to the $n = 6{,}443$ contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners inside the score window of T_{obs}: the rule of the plan, whose cohort is the $27{,}844$ contracts approved on any token from 2023-10-01, applied on the five tokens within that window. The rule counts every approving account, and of the addresses it lists those typed as user accounts are not in the cohort. W0 draws a cohort of its own by the same rule inside its own score window, so that no event after t_1 chooses its contracts. The owners of a cohort contract may all be user accounts, whose

approvals no graph reads, so a cohort contract need not appear in either graph and then scores 0 (Section 4.1.1 counts these contracts).

- Cohort events: The cohort rule, the spender lists and the labels count the events of every account around the addresses that the rule lists for a freeze, the events in which one of these addresses is owner or spender, sender or recipient. A spender that the rule does not list thus enters through an allowance that a listed address granted it, and its labels count only its events with listed addresses. The graphs hold every event between two contracts on the chain (Section 3.6).
- Exclusions: Mempool transactions, off-chain order books, centralized exchange accounts and bridges lie outside the data, and no attempt is made to merge addresses controlled by one operator into one entity. Tokens other than the five are left out, and native ether enters only as WETH.

3.3 Two-phase collection strategy

Collection and evaluation run as two phases, so that the tables, benchmarks and robustness runs can be rebuilt without cloud costs by anyone who has run the extraction.

- Phase 1 (reputation streams): Scan A copies the approval and transfer events from 1 October 2023 to 30 June 2026 into the study's project, and scan B those of July to September 2026 once the registration of W1 is committed. One query lists the spenders approved by at least three owners, the code check fixes the plan's cohort, and a further query extracts its ego events, from which the cohort, the spender list and the labels of each freeze are drawn on the five tokens. For the graphs, the end points of the five-token events are typed (Section 3.5.1), the owners' balances are computed (Section 3.5.2), and the events between two contracts are kept. At each anchor date the approvals of the score window are reduced to capped latest allowances and the transfers are aggregated per sender–recipient pair (Section 3.6), with amounts in US dollars; the proxies and the importance measures are computed alongside. These tables, of

which a last query checks 64 invariants, are the only inputs to graph construction.

- Phase 2 (local evaluation): From the downloaded Phase 1 tables a local machine computes the scores, the alignment statistics, the crawls of the discovery-order test, the holdouts, the benchmarks and the robustness runs, without querying BigQuery.
- Cost guardrails: Every query is first run as a dry run, and a query whose estimate would take the spending past the study's budget of USD 60 (USD 50 in the plan's configuration) is refused, and the jobs of the main analysis share a cap of their own within that budget.[1] Every job is written to a cost ledger with its billed bytes and cost (Appendix 8.1).

3.4 Pipeline architecture

Figure 3.2 traces the whole pipeline, from public logs to the research tables. The remote steps write tables in the study's BigQuery project, into which the account types read over RPC are loaded, and the downloaded files are never changed.

3.5 Event decoding

Decoding turns each log, whose topics and data carry no field names, into a typed record; an error here would corrupt both graphs silently.

- Log anatomy: Each log records the emitting contract, a sequence of topics and an ABI-encoded data blob; the first topic is the keccak-256 hash of the event signature, and the indexed parameters follow (Vogelsteller & Buterin, 2015). ERC-721 tokens emit `Transfer` and `Approval` under the same signatures but put the token ID in a fourth topic, so the condition of exactly three topics keeps them out. SQL filters by topic and decodes the ERC-20 fields.
- ERC-20 `Approval`: Its topic hash is

  `0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925`,

[1] This budget is enforced by the study's own pipeline; BigQuery does not impose it.

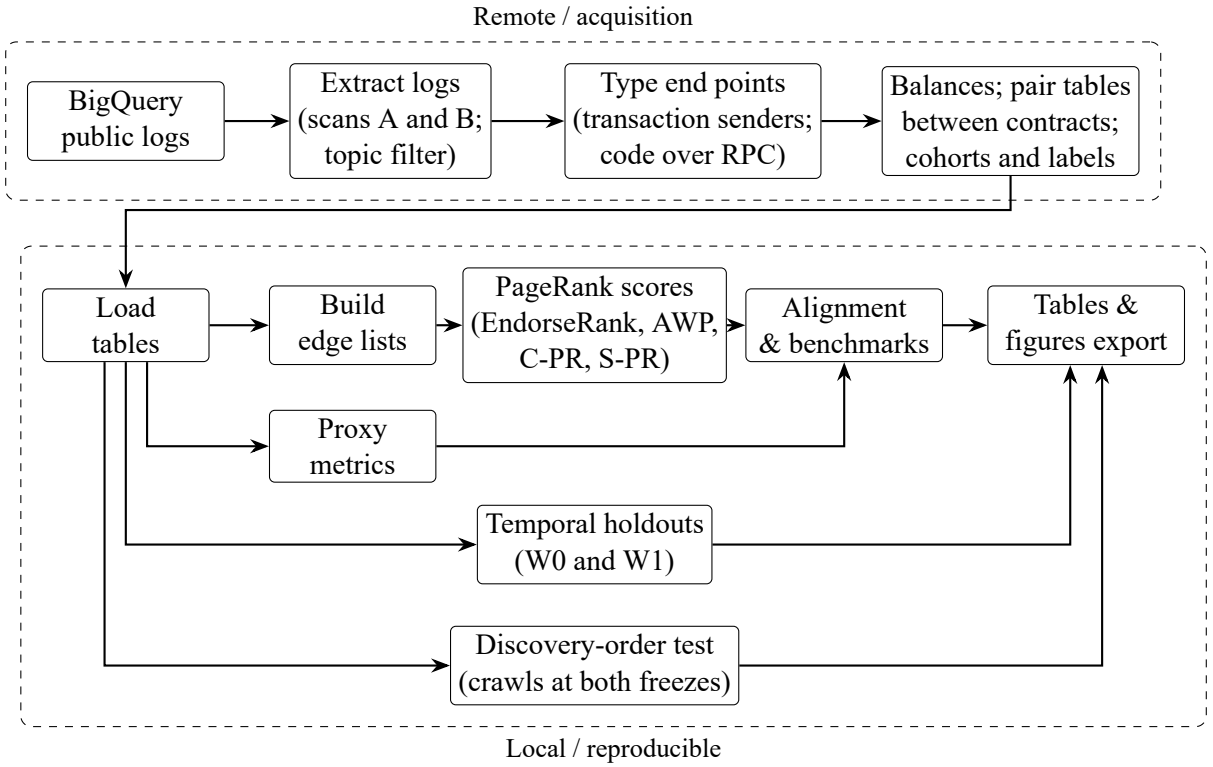


Figure 3.2: Pipeline overview. Approval and transfer logs are copied remotely in two scans; the end points of the five-token events are typed, the owners' balances are computed, and the events between contracts are aggregated into pair tables. Locally the pair tables become edge lists, from which the scores are computed; the alignment and benchmark calculations, the temporal holdouts and the crawls of the discovery-order test follow, and the tables and figures are exported last.

the keccak-256 hash of `Approval(address,address,uint256)`. Owner and spender occupy Topics 1 and 2, and the allowance is decoded from the data field in the token's base units and valued in US dollars (Section 3.5.3). An allowance of zero is kept, since it resets the allowance in force.

- ERC-20 `Transfer`: Its topic hash is

  `0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef`,

  the keccak-256 hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed, and the amount is valued in US dollars at the price of its day. Zero-value transfers are not kept. A mint or a burn names the zero address, which is neither a contract nor a user

account: it forms no edge and counts only in the balances (Section 3.5.2). Self-transfers stay in the events table but are left out of the inbound-counterparty computations (Section 3.10), since they reveal no counterparty.

- Address normalization and units: Addresses are lower-cased before the join. Amounts are the `uint256` value as a double-precision float, divided by ten to the power of the token's decimals and multiplied by its US-dollar price, so an unlimited allowance, $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units, is a very large dollar amount until the balance cap bounds it (Section 3.6).

3.5.1 Account types

An address other than the zero address that is an end point of a five-token event of the typing window is typed by four rules, applied in this order (Appendix 8.1.3; Table 8.4). (1) An address that sent a transaction in any block before 1 October 2026 is an EOA, and so a user account, since an address with deployed code cannot send one (Feist et al., 2021). (2) Every other address that could end an edge between contracts, that is one with a counterparty that is neither a transaction sender nor the zero address, and every member of a cohort or spender list, is read with `eth_getCode` at block $513{,}216{,}506$; no code, or an EIP-7702 delegation designator, means an EOA. (3) An address with other code is a smart account, and so a user account, if at the same block it answers as a Safe, with a threshold and owners, or as an ERC-4337 account, with a canonical EntryPoint contract and, in its code, the selector of `validateUserOp` and no paymaster selector. Two lists of code families extend this test to accounts whose proxy hides that code or whose EntryPoint getter has another name; they were drawn up from the code of the families alone, without reading a label, a cohort or spender membership or a score of the addresses they move. (4) Any other address with code is a contract, unless it is a system address (a precompile), which is never a node. Section 3.15 states the limits of these rules.

3.5.2 Balances

The balance that an owner held of a token at a freeze is reconstructed from the `Transfer` logs as the net flow of the token to the owner up to the freeze, mints and burns included: the public logs before scan A, scan A and the five-token events of the typing window. A negative value and a value below 10^{-6} token are set to zero. Balances are computed at t_1 and at T_{obs} for every (token, owner) pair of a five-token approval from 1 April 2025 to T_{obs}. As a check, $1{,}500$ sampled pairs are compared with the token's `balanceOf` at the last block of 30 June 2026; the pipeline stops unless 98% of them match. The check passed (Section 4.1).

3.5.3 Tokens, prices and the definitions of the main analysis

The analysis plan fixed amounts in raw base units, every ERC-20 token and $b = 1$ in Equation (3.4), on graphs that hold every account of the cohort's events. There the amount plays almost no part in EndorseRank or AWP, since $V(z)$ is within 10^{-4} of 1 for almost every event, the layers of C-PR add base units across tokens with different decimals, and any token contract can emit logs that name an owner or a sender that took no part.

After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. It is the main analysis of the thesis, called the contract-graph analysis where it stands beside the registered one.

Of the 133 tokens that Chainlink's directory of data feeds maps on Arbitrum One, 64 have a market-price feed, for the token or the asset it represents, with a low or medium market-risk category (Chainlink, n.d.), and 46 of these have a daily price at DefiLlama (DefiLlama, n.d.); the analysis keeps the five with the most rows among the events of the plan's cohort: WETH, USDC, USDT0, WBTC and ARB. A transfer is valued at the DefiLlama daily close of its UTC day, and a latest

allowance and a balance at the price of the anchor's day. The constants are set on the W0 score window of the contract graph, so that no event after t_1 enters them, and rounded to two significant digits. The slope of V is $b = \ln 3/m$, so that an amount of m dollars weighs one half: $m_T = 160$ dollars is the median transfer between contracts, and $m_A = 0.071$ dollars the median positive capped allowance between contracts at t_1, unlimited allowances included. $U = 16{,}000$ dollars, the 99th percentile of the same transfers, is the most that one allowance counts in the allowance layer of C-PR, and $U_{\text{user}} = 31{,}000$ dollars, the 99th percentile of the transfers from user accounts to contracts, the most that one user account adds to the importance of one contract.

Rule A and two of rule B's registered criteria are applied to the scores of the contract graph with the thresholds of the plan and the registration (Sections 3.8 and 3.8.1); the verdicts are a re-analysis and not registered tests. Appendix 8.6 sets the registered analysis beside the main one, and Appendices 8.4.2 and 8.4.3 give the tokens, the price checks, the account types and the balance cap.

3.6 Graph construction

EndorseRank and AWP weight their edges the same way and run the same PageRank solver, Equation (3.1); they differ in the events that make the edges and in where the walk restarts. An edge needs a contract at both ends, and the graph of a freeze holds every such edge of its score window, whatever cohort its ends belong to; a contract that ends no edge of positive weight in a method's graph scores 0 under that method.

Edges in EndorseRank stand for endorsements and edges in AWP for transfers (Section 2.8). Both methods weight an event by its age and pass its US-dollar amount through a bounded function, so that one event counts at most once.

3.6.1 AWP transfer graph

AWP is the method of T. Do et al. (2023). A transfer e between two contracts worth x_e US dollars, made Δt_e days before the score window closes at its anchor (T_{obs}, or t_1 in W0), enters the edge

from its sender to its receiver with the weight $\sigma(\Delta t_e)\, V(x_e)$. The time weight is the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp\big(k(\Delta t - t_0)\big)}, \qquad k = 0.01, \quad t_0 = 180 \text{ days}, \tag{3.3}$$

and the value weight is

$$V(z) = \frac{2}{1 + e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \tag{3.4}$$

which is 0 at $z = 0$, $1/2$ at the median amount $z = m$ and rises towards 1 as the amount grows. For transfers $b = b_T = \ln 3/m_T$ (Section 3.5.3). The edge weights are sums over transfers:

$$w_T(u, v) = \sum_{e:\, u \to v} \sigma(\Delta t_e)\, V(x_e). \tag{3.5}$$

The sum runs over every transfer from u to v in the window, whatever the token. A self-transfer ($u = v$) is kept and becomes a self-loop on u. The walk restarts at a contract in proportion to its activeness,

$$X_u = \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e), \tag{3.6}$$

and the restart distribution of Equation (3.1) is $\pi_u = X_u / \sum_{v \in V_T} X_v$, and it also takes the mass of the dangling nodes. A contract that never sends to another contract has $X_u = 0$ and receives no restart mass, so its score comes from its in-edges alone.

The paper scores Ethereum transactions and gives no parameter values; its time weight is the logistic function of Equation (3.3) written in calendar time and not in age, and k, t_0 and b are this study's choices, as is the restriction of the nodes to contracts. A transfer of twice the median amount weighs 0.8. Within a score window the decay discounts older events strongly (Figure 3.3): the weight falls from $\sigma(0) = 0.858$ through $\sigma(182) = 0.495$ to $\sigma(365) = 0.136$ for an event from the first day of the window.

Algorithm 1 states the construction and scoring steps formally.

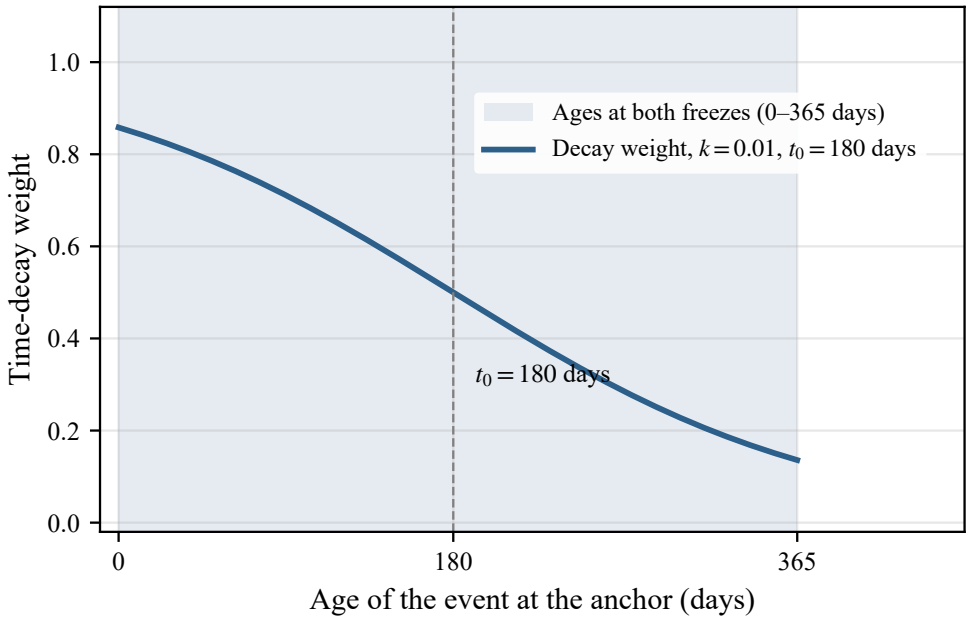


Figure 3.3: Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). The horizontal axis shows the age of an event in days before the anchor, $T_{\text{obs}} = 30$ June 2026 or, in W0, $t_1 = 31$ March 2026, and the vertical axis shows the time weight of a transfer or an approval. The shaded band marks the ages that occur at either freeze (0 to 365 days), over which the weight falls from 0.86 to 0.14.

Algorithm 1 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$ between contracts in US dollars, evaluated contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_T, d, \varepsilon, I_{\max}$

Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$

$E \leftarrow \emptyset$; $X_u \leftarrow 0$ for every contract u

for each transfer $e = (u, v, x_e, t_e)$ in $\mathcal{T}$ **do**

  $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days

  accumulate $\sigma(\Delta t)\, V(x_e)$ on edge $(u \to v)$ ▷ Equations (3.3)–(3.5); $u = v$ gives a self-loop

end for

for each sender u **do**

  $X_u \leftarrow \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e)$ ▷ Equation (3.6)

end for

remove non-positive edges

$\pi_u \leftarrow X_u / \sum_v X_v$ for the nodes u of the retained edges

$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)

return $\mathbf{s}_{\text{AWP}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

3.6.2 EndorseRank allowance graph

EndorseRank puts AWP's edge weights on the allowance graph. Let $\mathcal{A}$ denote the set of latest allowances between contracts after preprocessing. For each (token, owner–spender) triple only the most recent `Approval` event inside the score window is kept, in (block number, log index) order, and a zero allowance deletes the edge. The surviving allowance $a^\star(u, v, \text{token})$ counts up to the balance $\text{bal}(u, \text{token})$ of its owner at the freeze (Section 3.5.2), both in US dollars at the price of the freeze day, so the capped allowance is $\hat{a} = \min(a^\star, \text{bal})$. It was set $\Delta t^\star(u, v, \text{token})$ days before T_{obs} and enters the edge as one event of that age and of the capped amount:

$$w_A(u, v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)\, V\big(\hat{a}(u, v, \text{token})\big), \tag{3.7}$$

with σ, V, k and t_0 as for AWP and $b = b_A = \ln 3/m_A$. A directed edge $u \to v$ means that owner contract u endorses spender contract v. The spender can move at most the owner's balance, whatever the allowance says, so the capped amount is what the endorsement puts at stake; an owner that holds none of the token at the freeze gives no edge, however much passed through it in the window. The balance and not the allowance sets most capped amounts (Table 8.5), so m_A is small: a capped allowance of a dollar or more enters with its full time weight to four decimals, and V separates only the smaller ones.

The walk restarts uniformly over the contracts of the graph. AWP's restart rule, carried over to allowances, restarts the walk at owner u in proportion to $X_u^A = \sum_{v \neq u} \max_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)$, summed over the spenders v of u's positive capped allowances. That rewards granting endorsements, while EndorseRank is meant to rank the contracts that receive them. The analysis plan fixed uniform restarts and named this variant, EndorseRank with activity restarts, as one to report next to EndorseRank (Table 4.10).

The weight keeps the capped amount and the time of the latest approval; it is not the allowance in force at the end of the window. An allowance last set before the score window began is not

an edge, even if it is still in force, so the allowance graph holds the authorizations that owner contracts granted or renewed in the year before the freeze. An owner who approves an allowance-manager contract such as Uniswap's Permit2 appears only with an edge to that contract, not to the application that later spends through it. Some tokens also emit `Approval` from inside `transferFrom` with the reduced allowance. For them the latest record can be the remainder after the spender's last use, dated at that spend, so the time weight partly records use, an allowance spent in full ends at zero and gives no edge, and an authorization granted before the score window and used in a label window can appear as a new approval pair. How often the latest record is such a remainder was not measured.

Algorithm 2 lists the steps.

Algorithm 2 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$ between contracts in US dollars, owner balances bal, evaluated contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_A, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$

$E \leftarrow \emptyset$
for each latest allowance $(u, v, \text{token}, a^\star, t^\star)$ in $\mathcal{A}$ **do**
    $\Delta t^\star \leftarrow (T_{\text{obs}} - t^\star)$ in days; $\hat{a} \leftarrow \min\big(a^\star, \text{bal}(u, \text{token})\big)$
    accumulate $\sigma(\Delta t^\star)\, V(\hat{a})$ on edge $(u \to v)$ ▷ Equation (3.7)
end for
remove non-positive edges
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{ER}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

Allowance edges between contracts are far fewer than transfer edges (Table 4.1); since one PageRank iteration costs $O(|E|)$, this sparsity is the structural basis of H3.

Edge lists are stored as (from, to, weight) triples, with duplicate edges merged by summing their weights.

3.6.3 Coupled operator over both layers (C-PR) and the seeded ablation (S-PR)

EndorseRank and AWP each walk on a single edge type, so each ranks a contract by one kind of act: the spending rights it is granted or the payments it receives. C-PR walks both edge types of

the same contracts in one chain. The analysis plan fixed the coupled operator on two layers that weight each event by its amount; the main analysis values the amounts in US dollars and caps each allowance at its owner's balance:

$$\tilde{w}_A(u,v) = \sum_{\text{token}} \min\big(\hat{a}(u,v,\text{token}), U\big), \qquad \tilde{w}_T(u,v) = \sum_{e:\, u\to v} x_e\, \sigma(\Delta t_e). \tag{3.8}$$

The allowance layer has the edges of EndorseRank and the transfer layer those of AWP, apart from the few pairs whose value weight is zero in floating point (Section 4.1.1), which only the layers of C-PR keep; otherwise only the weights differ. Neither layer passes an amount through V, and only the transfers decay with age. Both rules for C-PR were fixed on these layers in raw base units. The two layers form one graph on the union node set $V = V_A \cup V_T$, with transition matrices P_A and P_T from Equation (3.2). For node u the coupled row is a convex mixture of the two layer rows, renormalized over the layers in which u has out-edges:

$$\begin{aligned} P^{(\lambda)}_{u,\cdot} &= \alpha_A(u)\, P_{A,u,\cdot} + \alpha_T(u)\, P_{T,u,\cdot}, \\ \alpha_A(u) &= \frac{\lambda\, \mathbf{1}[o_A(u) > 0]}{D_\lambda(u)}, \qquad \alpha_T(u) = \frac{(1-\lambda)\, \mathbf{1}[o_T(u) > 0]}{D_\lambda(u)}, \end{aligned} \tag{3.9}$$

where $D_\lambda(u) = \lambda\, \mathbf{1}[o_A(u) > 0] + (1-\lambda)\, \mathbf{1}[o_T(u) > 0]$, and $o_A(u)$ and $o_T(u)$ are the out-strengths of u in the two layers. A layer of weight 0 is left out with its edges and nodes, and a node with $D_\lambda(u) = 0$ is dangling, its mass handled by the third term of Equation (3.1). Teleportation is uniform over the contracts of V, $d = 0.85$, and $\mathbf{s}_{\text{CPR}}$ is the PageRank vector of Equation (3.1) with $P = P^{(\lambda)}$, under the solver settings of the single-layer methods. Figure 3.4 shows one step of the walk, and Algorithm 3 lists the construction and scoring steps.

At $\lambda = 1$ the operator walks the allowance layer alone and at $\lambda = 0$ the transfer layer alone; we write these walks C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). They are the comparators of both rules for C-PR (RQ4 and RQ5; Sections 3.8 and 3.8.1). The family is not continuous at the end points: for $0 < \lambda < 1$ a node with out-edges in only one layer follows it whatever λ is, whereas at $\lambda = 1$ a node

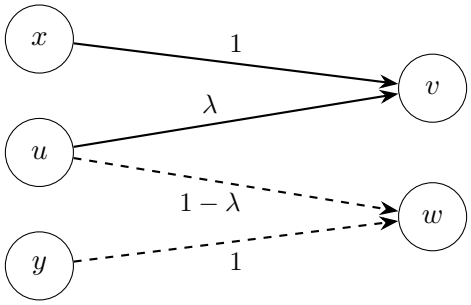


Figure 3.4: One step of the coupled walk C-PR (Equation (3.9)); solid arrows are allowance edges and dashed arrows transfer edges, labeled with their transition probabilities before damping. The contract u has out-edges in both layers and mixes them with weights λ and $1-\lambda$; x and y have out-edges in one layer only and follow it whatever λ is. With probability $1-d$ the walk restarts at a uniformly chosen contract.

Algorithm 3 C-PR graph construction and scoring

Require: Capped latest allowances $\mathcal{A}$ and transfers $\mathcal{T}$ between contracts in US dollars, evaluated contracts $\mathcal{W}$, observation end T_{obs}, cap U, weight λ, parameters $k, t_0, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{CPR}}$ on $\mathcal{W}$

allowance layer: accumulate $\min(\hat{a}, U)$ on $(u \to v)$ for each latest allowance in $\mathcal{A}$ ▷ Equation (3.8)
transfer layer: accumulate $x_e\,\sigma(T_{\text{obs}} - t_e)$ on $(u \to v)$ for each transfer in $\mathcal{T}$
remove non-positive edges; leave out a layer of weight 0
$V \leftarrow V_A \cup V_T$; $P_A, P_T \leftarrow$ row-normalized layers ▷ Equation (3.2)
for each node $u \in V$ with $D_\lambda(u) > 0$ **do** ▷ the others are dangling
    $P^{(\lambda)}_{u,\cdot} \leftarrow \alpha_A(u)\,P_{A,u,\cdot} + \alpha_T(u)\,P_{T,u,\cdot}$ ▷ Equation (3.9)
end for
$\mathbf{s} \leftarrow \text{WeightedPageRank}(P^{(\lambda)}, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equation (3.1)
return $\mathbf{s}_{\text{CPR}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

whose out-edges are all transfers is dangling or absent, and the same holds at the other end. The values $\lambda = 0.5$ (primary) and $\lambda \in \{0.25, 0.75\}$ (sensitivity) were fixed in the plan's configuration. Without its two caps the allowance layer would be very uneven: about two in three of the positive latest allowances between contracts are unlimited (Table 8.5), and each would take almost the whole row of its owner. The balance cap makes every allowance finite, and U keeps one large balance from taking the row. EndorseRank and AWP bound every amount through V.

In the S-PR ablation the transfer layer is walked, but the restart vector $\boldsymbol{\pi}$ of Equation (3.1) is not uniform: it is the score of C-PR ($\lambda = 1$), restricted to V_T and renormalized over it. Let $\mathbf{r}_A$ be the

PageRank vector of the allowance layer walked alone on V_A, extended by $r_{A,u} = 0$ for $u \notin V_A$. Then

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}} \qquad (u \in V_T).$$

If no node of V_T lies in V_A, the denominator is zero and $\boldsymbol{\pi}$ falls back to the uniform distribution on V_T. The result is the TrustRank construction (Gyöngyi et al., 2004), but with allowance-derived seeds in place of hand-labeled ones. A comparison of S-PR with C-PR separates what the allowance edge contributes as a propagation medium from what it contributes as a seed. Like AWP and C-PR ($\lambda = 0$), however, S-PR scores zero every contract without a transfer edge, about three spenders in ten at each freeze (Table 4.2). Most of them end no edge at all, but about 3% of each spender cohort has an allowance edge only, so a contrast of a transfer-only walk with C-PR, F1 of rule B included, mixes ranking with coverage.

Because C-PR walks on $E_A \cup E_T$, one of its iterations costs as much as an iteration on each layer, and its runtime is expected to be at or above AWP's; the speed result of Claim C6 is therefore EndorseRank's alone. S-PR needs one solve on each layer.

3.7 Discovery-order test

The discovery-order test repeats on the contract graph the crawl-ordering experiment of Cho et al. (1998) (Section 2.6). Its question is that of RQ2: ordered by C-PR computed on the contracts read so far, does a crawl from a random contract of the giant strongly connected component reach the contracts with the most user-account demand sooner than a breadth-first crawl and sooner than a crawl ordered by the number of contract backlinks? The test was defined after the registered results were known and decides no rule of the plan or the registration; its protocol is `experiments/docs/discovery_protocol.md` in the repository.

A page is a contract and a link an edge of the graph G of one freeze, in either layer; T_{obs} is the primary freeze and t_1 the secondary one. A crawl starts at a seed and visits one contract at a time, knows only the edges that its visits have revealed, and chooses the next visit from its frontier,

the contracts discovered and not yet visited. The union graph holds the distinct ordered pairs of different contracts with an edge in either layer. In the `out` mode a visit reveals the out-edges of the contract, the approvals it granted and the transfers it sent, and the universe $\mathcal{U}$ is the giant strongly connected component of the union graph with every contract reachable from it, SCC $\cup$ OUT in the bow-tie decomposition of Broder et al. (2000). In the `both` mode a visit reveals the in-edges as well, and $\mathcal{U}$ is the weakly connected component that holds the giant SCC (WCC). The protocol makes `both` the primary mode if SCC $\cup$ OUT holds fewer than half the contracts of the WCC in the structure pilot below, as it does, and `out` is then a sensitivity (Table 8.7).

The seeds are the 20 contracts of the giant SCC with the lowest SHA256 hash of a fixed string and the address. An ordering other than the random one scores every frontier contract: after K visits the scores are recomputed, and the $\max(1, \lceil \rho K \rceil)$ highest-scored frontier contracts are visited before the next recompute, ties going to the earlier discovery and then to the lower address. A prime marks what is computed on the contracts read so far. The orderings are breadth-first (BFS), which follows the order of discovery; random; three counts of the visited contracts with an edge to the frontier contract, the raw degrees that the walks smooth (IB'_A on allowance edges, IB'_T on transfer edges and IB'_U on either, the contract backlinks); and the walks of Section 3.6 on the graph seen so far (EndorseRank′, EndorseRank-activity′, AWP′, C-PR′ at $\lambda = 0.5$, 0.25 and 0.75, its two layers and S-PR′), each recompute starting from the previous vector, a warm start that is part of the method. The ideal order and the orders by C-PR and by union in-degree of the full graph are references that are not crawls, and the normalized in-strength δ', the depth-one closed form of the walk, is a diagnostic outside every comparison.

Importance is user-account demand, which no ordering reads. For a contract c of $\mathcal{U}$ and the score window of the freeze, $I_T(c) = \sum_j \min(x_{j\to c}, U_{\text{user}})$ and $I_A(c) = \sum_j \min\big(\hat{a}(j,c), U_{\text{user}}\big)$ with $\hat{a}(j,c) = \sum_{\text{token}} \hat{a}(j,c,\text{token})$, where j runs over the user accounts, $x_{j\to c}$ is the US dollars that j transferred to c, and $\hat{a}(j,c,\text{token})$ is the latest allowance of j to c on a token capped at the balance of j, as in Equation (3.7). The numbers of accounts with at least one dollar, N_T and N_A, are secondary. Importance in the manner of Cho et al., here the union in-degree IB_U and C-PR

(IR) of the full graph, is measured on the edges that the orderings read, so agreement with it is an intended-construct check, reported second.

The hot set $H_q(I)$ holds the contracts of $\mathcal{U}$ whose importance is positive and among the $\lceil q|\mathcal{U}| \rceil$ largest, ties included. With t_h the visit at which a crawl reaches hot contract h, AUC_x is the mean over the hot contracts and the seeds of $\max(0, 1 - t_h/\lceil x|\mathcal{U}| \rceil)$, the normalized area under the curve of the share of the hot set visited during the first $x|\mathcal{U}|$ visits; the random reference curve, $K/|\mathcal{U}|$ after K visits, gives $x/2$. The primary values are $x = 10\%$ and $q = 1\%$. Intervals come from a paired bootstrap over the hot contracts ($B = 400$, seed 42, one index matrix for every ordering).

The decision is read at T_{obs} in the primary mode, on $H_q(I_T)$ and on $H_q(I_A)$, from two contrasts of $\mathrm{AUC}_{10\%}$: D_1, C-PR′ minus BFS, and D_2, C-PR′ minus IB'_U. Section 6 of the protocol words the verdict on each importance:

- V1, both intervals above 0: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most [user-account inflow | user-authorised value] sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks."
- V2, the D_1 interval above 0 and the D_2 interval not above 0: "… sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks."
- V3, the D_1 interval not above 0: "… did not reach the contracts of the universe with the most [user-account inflow | user-authorised value] sooner than a breadth-first crawl."

An interval below 0 adds a sentence naming the ordering that was faster. "The answer to the question of section 1 is the weaker of the two verdicts (V2 is weaker than V1, and V3 than V2), so it is yes only when both are V1"; the question of the protocol's first section is that of RQ2. At t_1 and in the other mode the same contrasts are reported without a verdict.

Outside the decision the protocol reports, in a fixed order: each walk′ against the count it smooths; contrasts among the walks, first C-PR′ against C-PR′ ($\lambda = 0$) and against AWP′, which show what the allowance layer adds; the other values of λ; the hot sets of IB_U and IR, where C-PR′ minus IB'_U

on $H_q(\mathrm{IB}_U)$ repeats the comparison of Cho et al.; sensitivities, the other reveal mode among them; and the freeze t_1, which shares nine months of its score window with T_{obs}.

A structure pilot on the t_1 graph, which reads the pairs of each layer without weights and computes no importance, score or crawl, sets three values by rules of the protocol, for both freezes: the primary mode, $\rho = 0.02$ (the smallest of 0.01, 0.02 and 0.05 with a projected run time of at most twelve hours) and $q = 1\%$. The pilot also raised the stop flag that the protocol sets on a thin allowance layer, since 0.54% of the contracts of SCC $\cup$ OUT have a capped allowance in-edge against a threshold of 5%, and the test went on by the author's decision, which the protocol records with its reasons, because it is the only direct test of the question.

3.8 Temporal holdout protocol

In the holdout W0, scores use the events between contracts of the W0 score window, from 1 April 2025 up to t_1 = 31 March 2026, 23:59:59 UTC. Labels are counted on events from 1 April 2026, 00:00:00 UTC, up to $t_2 =$ 30 June 2026, 23:59:59 UTC. The cohort consists of the $n = 7{,}864$ contracts among the spenders that hold at least one positive latest allowance of one of the five tokens, set inside the window, at t_1, in the events around the addresses listed for W0 (Section 3.2); these are listed from the approvals of the same window, so no event after t_1 chooses the spenders. A spender that ends no edge of the contract graph stays in the cohort and scores zero under every score and baseline. Every score in the holdout table (EndorseRank, AWP, C-PR at the three values of λ and at its two end points, S-PR and four raw counts) is computed from the contract graph of this window alone, with the time decay anchored at t_1, and so is EndorseRank with activity restarts (Table 4.10).

Two labels are counted per spender, one for each construct, so that each single-layer method is tested on its own construct and the hybrids on both:

1. New approval pairs: the count of owner–spender pairs, attributed to the spender, that receive a positive approval in the label window but were absent from the latest-positive snapshot at

t_1. This label measures the endorsement construct.

2. New transfer senders: the count of distinct addresses that send the spender a transfer in the label window without having sent it one between the start of the score window (1 April 2025) and t_1, self-transfers excluded. It measures the flow construct.

Both labels count every account in the events around the listed addresses, and most of what they count comes from user accounts (Table 8.6), whose events no score reads: the labels ask whether the standing of a contract among contracts predicts later demand from all accounts. The baselines are raw counts on the contract graph at t_1: each spender's in-approve degree (the distinct owner contracts with a positive capped allowance to it), its in-approve value (Equation (3.13)), its transfer in-degree (the distinct sender contracts other than itself) and its union in-degree (the distinct other contracts with an edge to it in either layer). Each single-layer PageRank smooths one of the first three and C-PR the fourth. The increment of a PageRank over its own raw count answers RQ3; the PageRank coefficient on its own does not.

Every difference between two coefficients on the same label is a paired difference on the resamples of Section 3.11. The contrasts fixed in the plan are the two of rule A below; C-PR against the union in-degree was named when the main analysis was defined and belongs to no rule. Every other contrast in this window was computed after the labels were known; the contrasts of C-PR with every other score are in Table 4.22, and the differences between EndorseRank and AWP in Appendix 8.5.

Rule A, fixed in the analysis plan, judges C-PR against its own layers walked alone and has two parts. The primary part uses W0: C-PR at $\lambda = 0.5$ should do no worse than C-PR ($\lambda = 1$) on new approval pairs (the interval of $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=1}$ contains zero or lies above it) and should be better than C-PR ($\lambda = 0$) on new transfer senders (the interval of $\tau_{\text{CPR}} - \tau_{\lambda=0}$ lies above zero). The secondary part uses the same window and the matched cohort. Its condition is that C-PR's point estimate lies within the 95% interval of the better of the two layers on the transfer family and on the allowance family, and that C-PR exceeds both layers on the Sybil-stability family. The paired

contrasts of Table 4.19 are reported beside this rule as a stricter check; they do not decide it. If its primary part fails, RQ4, whether C-PR beats a walk on either layer alone, receives a negative answer. RQ5, whether C-PR improves on a walk over the transfers alone, is not covered by rule A; two criteria of rule B test it in W1 (Section 3.8.1). Once labels have been observed, λ is not tuned. The pipeline also counts, over the label window, revocations, the pairs held at t_1 that receive a zero approval (for some tokens an allowance spent in full, Section 3.6.2), and an approval-then-outbound drain heuristic. Neither is offered as supporting evidence (Section 4.7).

Two descriptions beside the rules decide none of them: the main comparisons are repeated on the spenders inside the union graph, outside which more than a quarter of each spender cohort ties at zero under every score (Tables 4.2 and 4.23), and each label is split by the class of the account it counts (Table 8.6).

3.8.1 Registered window W1: July to September 2026

The W0 holdout also showed something that no rule had asked about: against its transfer layer walked alone, C-PR ranked the spenders that later gained new approval pairs higher, with an interval above zero, while it was worse on new transfer senders (in the registered analysis; Appendix 8.6). Noticed only after the labels were in, this counts as a hypothesis and not as a finding, and it is tested again on a window that the study had not extracted when the test was fixed. The registration document, written after the W0 results were known, and the configuration, with the status of the window set to registered, were committed to the authors' Git repository, and scan B, the first query to read logs dated after 30 June 2026, ran after that commit, an order that the extraction script enforces (Appendix 8.1.1).

The method is unchanged: C-PR at $\lambda = 0.5$, as fixed in the plan, with the two sensitivity values and S-PR, and the solver settings, label definitions and paired bootstrap of W0. Scores are frozen at T_{obs}, so they are the scores of the same-window evaluation: in the registration, which belongs to the registered analysis, they use every extracted event from 1 October 2023 onwards on graphs that hold every account of the cohort's events, and in the main analysis the events between contracts of

the score window of T_{obs}. Labels are counted from 1 July to 30 September 2026.

The spender cohort is defined as in W0: the $n = 7{,}915$ contracts among the spenders holding at least one positive latest allowance at the freeze, with the two labels of W0. It holds about five in six of the W0 spenders, so W1 is a later label window on largely the same contracts.

Rule B compares C-PR with its transfer layer walked alone, C-PR ($\lambda = 0$), through the paired difference $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=0}$. The thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds.

- F1, spenders, new approval pairs: the 95% interval of $\Delta\tau$ lies above zero;
- F2, spenders, new transfer senders: the lower end of the interval lies above $-\delta$.

If F1 or F2 fails, rule B is not met; if both hold, the reading is that both applied criteria hold, and no verdict is given on the rule as registered. The margin $\delta = 0.01$ is the half-width of the paired interval for the F2 contrast in W0 in the registered analysis, rounded up to two decimals, as the registration fixed it; in both analyses the spender cohorts resolve differences smaller than that (Table 8.16). Outside the decision the registration reports the primary part of rule A rerun on the new window (F6): C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs (F6a) and better than C-PR ($\lambda = 0$) on new transfer senders (F6b).

Two sensitivity analyses were registered in the same commit, before the study extracted any July–September log. Their spender parts are reported next to the two criteria, and neither enters the reading. The first repeats F1 and F2 with AWP (Section 3.6.1) in place of C-PR ($\lambda = 0$). The second repeats F1, F2 and F6 with the spenders that a graph leaves out kept as isolated nodes and not scored zero (Section 4.4), in the graph of every walk but S-PR. Other contrasts in W1, each PageRank against its own count and C-PR against every other score among them, were computed after the labels were known and are reported apart from the two criteria.

3.9 Construct-check design

Every score enters the same-window evaluation: EndorseRank, also with activity restarts, AWP, C-PR at its five values of λ and S-PR. For each score M and check c the pipeline reports Spearman's ρ and Kendall's τ between $\mathbf{s}_M$ and $\mathbf{p}_c$ with 95% bootstrap intervals (Section 3.11).

The same-window evaluation is organized around two contemporaneous families: transfer, the contract's inbound transfer activity (the proxies used to evaluate AWP (T. Do et al., 2023)), and allowance, its inbound approval activity as a spender. Both are graph-adjacent: each check is a local statistic, not a global score, of the events between contracts that make up G_{transfer} and G_{approve}. Alignment of AWP with the transfer checks, or of EndorseRank with the allowance checks, is an intended-construct check: it shows coherence within a construct and is not evidence of independent predictive power. A third family, the Sybil-adjusted activity summaries of Section 3.10.3, enters the secondary part of rule A, but no hypothesis rests on it.

The discovery-order test, whose importance comes from user accounts, and the temporal holdout, whose labels come after the freeze, do not rest on the edges that a score has read. Packin and Lev-Aretz (2024) warn that decentralized scoring can become opaque; publishing explicit formulas (Section 3.10) and keeping the same-window checks apart from the other two evaluations reduce that opacity.

3.10 Operational definitions of proxies

In what follows, $\mathcal{W}$ denotes the matched cohort, and every proxy is computed from the events between contracts in the score window of T_{obs}.

3.10.1 Transfer family

Write $\mathcal{T}_{\text{in}}(w)$ for the set of transfers that contract w receives from contracts in the window and u_e for the sender of transfer e. Self-transfers belong to $\mathcal{T}_{\text{in}}(w)$, so a contract that sends tokens to itself

counts itself as one sender, and amounts x_e are summed in US dollars across the five tokens. Define

$$\text{in_degree}(w) = \big|\{\, u_e : e \in \mathcal{T}_{\text{in}}(w) \,\}\big|, \tag{3.10}$$

$$\text{in_value}(w) = \sum_{e \in \mathcal{T}_{\text{in}}(w)} x_e. \tag{3.11}$$

Both are baselines defined in the AWP paper (T. Do et al., 2023). Unlike PageRank they are local and ignore the senders' reputations, so comparing AWP with them shows how far propagation, recency weighting and restarts at active senders move its ordering away from the raw inflow.

3.10.2 Allowance family

On the allowance side, w is the spender (owner $\rightarrow$ w). With $\hat{a}$ the capped latest allowances of Equation (3.7), define

$$\text{in_approve_degree}(w) = \big|\{\, u : \hat{a}(u, w, \cdot) > 0 \,\}\big|, \tag{3.12}$$

$$\text{in_approve_value}(w) = \sum_{u,\text{token}} \min\big(\hat{a}(u, w, \text{token}), U\big). \tag{3.13}$$

Like the transfer baselines, they are local and follow no multi-hop endorsement chain.

3.10.3 Sybil-adjusted stability family

The inbound counters leave out self-transfers ($u = w$). Let $N_{\text{tx}}(w)$ be the number of transfers that w receives from other contracts and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Tenure and active months use a second set of rows. Every transfer between contracts that involves a cohort contract is attributed to its sender if the sender is in the cohort, and otherwise to its recipient; a transfer between two cohort contracts thus counts only for the sender. We take $t_{\text{min}}(w)$ and $t_{\text{max}}(w)$ to be the earliest and latest timestamps among the rows attributed to w. $N_{\text{mon}}(w)$ is the larger of two counts of distinct calendar months: the months in which w received a transfer from another

contract, and the months of the rows attributed to w. Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \tag{3.14}$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad \text{(seconds to days)}, \tag{3.15}$$

$$\text{active_months}(w) = N_{\text{mon}}(w). \tag{3.16}$$

The counterparty ratio penalizes contracts that receive many transfers from few counterparties, a simple wash pattern. It does not penalize the usual Sybil pattern, in which many fresh contracts send one transfer each; a contract fed that way scores close to the maximum of one. Tenure and active months track persistence over the score window. These proxies are therefore weak Sybil checks, and colluding adversaries can satisfy all three.

3.11 Statistical tests

Concordance is measured with two rank correlations, as in the AWP evaluation protocol (T. Do et al., 2023), so that no conclusion hinges on one coefficient; both are invariant under monotonic re-scaling, and the few hubs of the heavy-tailed scores do not dominate them.

- Spearman's ρ: the Pearson correlation of the mid-ranks R_i and S_i of the raw score and the raw proxy value of contract i (Section 2.10).
- Kendall's τ: every reported value is the tie-corrected τ_b of Equation (2.6), computed on the raw scores and proxies. Without ties, $(1+\tau)/2$ is the probability that a randomly chosen pair is ordered the same way by both rankings. With ties that reading belongs to $\tau_a = (n_c - n_d)/n_0$ and not to τ_b. Ties are common: a contract outside a method's graph scores zero, so the scores on the allowance layer tie most pairs of the matched cohort and the other scores a smaller share (Section 4.4), and the raw counts and the labels tie as well. When the two sides of a paired contrast tie very different shares of pairs, τ_b gives the more tied side the smaller denominator, so the contrast can favor it under τ_b and not under τ_a; the contrasts, computed with τ_b alone,

are exposed to this.

- Family aggregation: A family is summarized by the plain average of the coefficients of its proxies. Conclusions are drawn from the confidence intervals, and no absolute τ cutoff is used as a decision criterion.
- Uncertainty: Sampling uncertainty is estimated with a paired bootstrap over contracts (Efron, 1979). The evaluated cohort is resampled with replacement $B = 400$ times (seed 42, fixed in the configuration), and one index matrix is reused for every method and proxy, so the resampled statistics are paired across methods. For each proxy τ, family mean and inter-method difference, the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval (DiCiccio & Efron, 1996). The percentile interval is first-order accurate at the sizes of the cohorts if contracts are independent; contracts that share owners or belong to one protocol are not, so the intervals may be too narrow. With $B = 400$ the Monte Carlo error of an interval end is about 0.13 times the bootstrap standard error. The discovery-order test resamples the hot contracts in the same way, after averaging the crawls of the 20 seeds for each hot contract (Section 3.7); hot contracts are not independent either and the seeds are few, so a bootstrap over the seeds is reported as a sensitivity.
- Difference tests: Every comparison is a paired contrast on the same resamples: we compute $\Delta\tau = \tau_A - \tau_B$ resample by resample and report the 95% percentile interval of its bootstrap distribution, and a contrast counts as a difference when its interval excludes zero. The intervals of the many contrasts computed after the labels were known are not adjusted for multiplicity, and a few would exclude zero by chance alone; a post hoc contrast resolved in one window only is therefore read as a hypothesis, and one resolved in the same direction in both windows as less exposed to chance. The contrasts that decide a rule were fixed before the study extracted the data they are judged on: those of rule A (Section 3.8) before any log, and F1 and F2 of rule B before the W1 logs; on the scores of the contract graph they are applied as a re-analysis (Section 3.5.3). C-PR against its transfer layer walked alone on new approval pairs in W0 belongs to no rule and is marked as exploratory wherever it appears. In the same

window, eight contrasts set C-PR and S-PR against whichever layer, walked alone, leads each family, and six set C-PR against AWP and EndorseRank; none of these fourteen was fixed in the plan, and two of them were computed afterwards from the saved scores. Out of window, the contrasts of C-PR with every other score and with the four raw counts (Table 4.22) were computed after the labels were known, as were the five same-window contrasts of EndorseRank and AWP in Appendix 8.5.

- Directionality and imputation: Each proxy is oriented so that a higher value means a better reputation. A contract absent from a proxy source or from a graph takes the value zero, under every score alike, so each correlation uses the whole cohort.
- Non-inferiority: F2 of rule B (Section 3.8.1) asks only that C-PR not fall behind its transfer layer walked alone by more than the registered margin $\delta = 0.01$. The test is weaker than a difference test and says nothing about superiority on that label.
- What is excluded: The study attempts no causal inference, trains no supervised models on separate training and testing sets, and reports no p-values.

3.12 Computational benchmark (Claim C6)

The benchmark of Claim C6 times the solver call that turns an edge list into scores; a scoring service that refreshes its scores periodically would amortize the one-time cost of building the edge lists.

- Timed component: The edge lists and AWP's restart weights are built before the timer starts. The timed call indexes the nodes, assembles the sparse weighted adjacency matrix, row-normalizes it, forms the restart vector and runs power iteration with $d = 0.85$, $\varepsilon = 10^{-8}$ and $I_{\max} = 300$. Matrix assembly and each iteration take time that grows with $|E|$ and $|V|$, and a deployment would repeat both at every update.
- Measurement protocol: Every configuration is measured in one session. An untimed warm-up run, on which alone peak memory is tracked, is followed by five timed runs, from which we report the mean and standard deviation of the run times and the iteration count, with the

nodes and edges of the input graph. A graph and setting that appear in more than one table are measured once (Appendix 8.1.4). Every runtime table lists edge counts, since edges drive the cost of an algorithm with $O(|E|)$ work per iteration (Langville & Meyer, 2006).

- Graphs: The main benchmark runs on the whole contract graph of T_{obs}, which holds every pair between contracts and not only the pairs of the matched cohort: the allowance graph, the transfer graph and, for C-PR, their union. For the growth of runtime with graph size, three deterministic subsamples of the cohort keep the pairs incident to the sampled contracts (Section 3.13.1).
- Fairness: Every algorithm runs with the same solver, tolerance, iteration cap and damping (outside the damping sweep), so differences in cost come from the graphs and from AWP's restart vector, which can change the number of iterations, and not from the implementation. The cost of two scores is compared through time ratios, each reported beside the corresponding edge ratio.

3.13 Scaling and robustness extensions

None of the analyses of this section supplies an alternative primary sample: alignment claims in Chapter 4 refer to the matched cohort unless stated otherwise.

3.13.1 Cohort subsamples for runtime scaling

The scaling stages are drawn from the matched cohort by deterministic subsampling: contracts are sorted by the SHA256 hash of the string made of the seed `contract-benchmark-v1`, a colon and the lower-cased address, and a stage is the first n contracts in that order, so each stage contains the one before it and the subsamples cannot drift between library versions the way pseudorandom draws do. We report three stages, $n = 1{,}000$, $2{,}000$ and $5{,}000$, and after them the main measurement on the whole contract graph.

Three features of this design limit what the experiment can show. A stage keeps the pairs of the

contract graph at T_{obs} incident to its contracts, those with at least one end among them, so $|E|$ grows with the stage but not in proportion to n, since a contract with many counterparties brings many pairs. The last row is not a fourth stage of this construction: it is the whole contract graph, which also holds the pairs between contracts outside the cohort, and the graph of the pairs incident to the full cohort ($n = 6{,}443$) has no timing of its own. And no row goes beyond the contract graph of T_{obs}, so the experiment says nothing about larger graphs.

3.13.2 Damping sensitivity

On the matched cohort the damping is varied over $d \in \{0.75, 0.85, 0.95\}$, with the graphs and proxies held fixed, to see whether alignment on the allowance and transfer families depends on the teleportation probability.

3.13.3 Value-slope sensitivity

For this check EndorseRank and AWP are recomputed at T_{obs} with the slope b of V multiplied and divided by ten, which moves the amount of half weight to a tenth and to ten times the median; the plan's check on twenty-token subgraphs belongs to the registered analysis.

3.13.4 Sample-definition sensitivity

At each scaling stage of Section 3.13.1, and for the full cohort, the family means of EndorseRank and AWP are recomputed on the stage's contracts, scored on the pairs incident to them; the full cohort too is scored here without the pairs between contracts outside it, on a graph that has no timing of its own. A contract's score changes because those pairs are missing, so the per-stage coefficients are a sensitivity check on the sample definition and not evidence of stability, and no intervals were computed for them. This analysis was not in the plan.

3.13.5 Uncapped allowance layer

One sensitivity analysis shows what the balance cap changes: it repeats the comparisons of every score and baseline on the W0 and W1 spender labels with an uncapped allowance layer, in which every positive latest allowance between contracts counts at $a^{\star}$ in place of $\hat{a}$ in Equations (3.7) and (3.8). It is reported next to the main comparisons (Section 4.5.5), decides no rule, and the discovery-order test does not use it.

3.14 Reproducibility

No table in Chapter 4 was transcribed by hand. Once the Phase 1 tables are on disk, the evaluation drivers recompute every check, every alignment statistic with its interval, every crawl, every holdout result and the benchmark, and write them to machine-readable summaries, from which `scripts/export_latex.py` exports the tables and figures. One versioned configuration file holds the windows, cohort threshold, event-topic IDs, damping, decay parameters, values of λ, bootstrap count and seed, benchmark seed and cost budget; an overlay for the main analysis (variant `contract12m`) sets the start of each score window, its budget and the settings of the typing, of the balance check and of the discovery-order test. `scripts/run_contract12m_pipeline.py` runs the remote steps in order, and it and `scripts/run_discovery.py`, which runs the crawls, refuse to run unless the documents and the code they rest on are committed without local changes.

The plan and the registration of W1 were committed to the authors' Git repository, which is private. Its folder for this study is published with its commit history in the public repository https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation, which also holds the documents, the code and the results of the main analysis; the file `HISTORY.md` maps the commit identifiers that the extraction manifest and the digest of the registered analysis record. There the plan and the registration are `experiments/docs/analysis_plan.md` and `experiments/docs/registration_w1.md`, and four files of `experiments/docs/` define the main analysis: `revision_price_weighting.md`, `revision_one_year_window.md`, `revision_contract_graph.md`,

with the lists of code families, and `discovery_protocol.md`. These files fix every element that could vary between runs: the subsampling of the cohort by SHA256 hashes, the solver settings, the bootstrap seed and the resample-index matrix of each cohort and hot set, the coupling weight λ and the holdout logic, the seeds, the schedule and the tie rule of the crawls, the pinned block of the account types and the extraction manifest of scan B. Appendix 8.1 names the commit of the summaries behind the tables of this thesis, the configuration file and the summary files with their digest.

3.15 Sampling considerations and missingness

The matched cohort is a constructed sample, not a random draw from all Arbitrum addresses. The same-window findings apply to the contracts that the cohort rule selects, whatever their transfer activity, and the holdout findings to the contract spenders in the events around them; user accounts and contracts with fewer approving owners are excluded by design (Section 3.2).

The cohorts are chosen by the approvals of every account and the graphs hold only the events between contracts, so about a third of each cohort and more than a quarter of each spender cohort end no edge and score zero under every score and baseline (Table 4.2); the comparisons inside the union graph set this effect apart (Table 4.23). The discovery-order test has a coverage of its own, the share of each importance that its universes hold (Table 8.8).

The typing rules have limits. A smart account that is neither a Safe nor an ERC-4337 account with a canonical EntryPoint stays a contract and a node: after the listed code families, 115,084 typed contracts still hold a `validateUserOp` selector and no paymaster selector, and no rule picks out the contracts that a service deploys for each user, payment or order, which, judged from their code, are several of the largest code families among the nodes (section 13 of `experiments/docs/revision_contract_graph.md`). The lists move whole code families, with the hidden code read for sampled addresses only, and every address is typed by its code at one block, nine days after the end of the data. An address that sent no transaction and has only transaction senders as counterparties

is not read: it cannot end an edge between contracts, and whatever it is, it lies outside every graph and every universe of the discovery-order test.

Data can be missing in a few ways. An amount that cannot be parsed is read as zero, which loses a transfer and resets an allowance. EndorseRank does not see approvals that emit no standard ERC-20 event, and AWP does not see transfers made off-chain or inside custodial accounts. A balance that changed without a `Transfer` log would escape the reconstruction of Section 3.5.2. The scores are thus biased toward on-chain, standard-token behavior, which is acceptable for an on-chain reputation study but matters once results are extrapolated to all contracts of a protocol.

3.16 Implementation notes

The PageRank code is written in Python, with a SciPy sparse transition operator and NumPy power iteration. Appendix 8.1.1 names the machine on which the benchmark ran. Randomness enters only through the bootstrap and the random ordering of the discovery-order test, whose seeds are fixed; given the downloaded tables and the configuration, evaluation is deterministic. Floating-point differences between platforms would be negligible against the three-decimal precision of τ.

3.17 Ethics

Only publicly available blockchain data are used in this research, and it involves no human subjects. The study was reviewed by the institutional ethics committee (Unisa SoC/CAES). Appendix 8.7 contains the ethics clearance certificate (reference [Unisa SoC/CAES ethics clearance ref: TBD], dated [TBD]).

- Privacy and anonymity: An address on Arbitrum One is a pseudonym and does not by itself reveal the real-world identity behind it (Packin & Lev-Aretz, 2024). Pseudonymity is not anonymity: a determined adversary could tie an address to a person through exchange deposits, online profiles or heuristics, and this work makes no such links. The scored addresses are contracts, and no result is reported for any user account. Our data collection is limited

to on-chain `Approval` and `Transfer` events, the addresses that sent a transaction and read-only calls for the code, the smart-account tests and the balance check, and we scraped no off-chain personally identifiable information (such as IP addresses, exchange KYC records, social media handles or deanonymization graphs). We did not cluster addresses into real-world entities, publish leaderboards of addresses likely to be personally identifiable, or try to join an address's transaction history with other databases to infer identity. The published results are aggregate measures only (correlation coefficients, runtimes, family means) and contain no individualized rankings of named participants.

- Data minimization: Scans A and B keep every ERC-20 `Approval` and `Transfer` event of their months in the study's cloud project. The tables downloaded for the main analysis hold only what graph construction and the checks require: pairs of contracts with their weights, sums over user accounts for each contract, the cohort lists with their labels, the class of each typed address and the balances of the check; none lists the approvals or transfers of a user account. The extracted tables are kept so that the research pipeline can be audited. They will not be released for the purpose of deanonymizing users.
- Data usage and compliance: Every query ran against public data hosted in the `bigquery-public-data.goog_blockchain_arbitrum_one_us` dataset and complied with the Google Cloud terms of service and rate limits (Section 3.3). Account types, smart-account tests, the balance check and token data were read with read-only calls to public RPC endpoints, and daily prices came from the public DefiLlama API. We sent no transactions and accessed no privileged or non-public indexing endpoints, and no call could interfere with the operation of the network.
- Fairness and transparency: Decentralized scoring raises concerns about fairness as well as opacity (Packin & Lev-Aretz, 2024). New, rarely used and unapproved contracts may be undervalued by graph-based reputation systems, EndorseRank and AWP among them: the allowance graph disadvantages an address that has received no spend authority, and the transfer graph favors addresses embedded in value-flow networks. On graphs of contracts, a contract

that only user accounts approve or pay ends no edge and scores zero. Chapter 5 discusses the limitations, among them cold-start effects and the Sybil susceptibility of approval graphs.

- Responsible disclosure and impact: The scores of this study may inform risk assessment in DeFi, but they are not financial advice or a consumer reporting product. Lending, trading and compliance decisions still need human oversight. Chapter 5 discusses adversarial manipulation, for example Sybil attacks on G_{approve} or wash transfers on G_{transfer}. Any future operational deployment would require additional governance, appeal mechanisms and fairness monitoring, all of which lie beyond the scope of this research.
- Summary of methodological commitments: The plan and the registration fixed their rules before the study extracted the data they are judged on, Section 3.5.3 gives the status of the definitions of the main analysis, and each PageRank is set against the raw count it smooths.

Chapter 4 reports what this protocol yields for Arbitrum One.